

Stomach, Reflux and Helicobacter

Gastroesophageal and esophagus reflux of Barrett, hiatus hernia, peptic ulcer, Helicobacter pylori response, gastritis, gastric cancer and proton pump inhibitors

Sérgio F · 02-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

Contents

Section	Page
1. Summary	3
2. How to read this report	4
3. Polygenic scores	5
4. Key variants by topic	6
5. GWAS loci burden	9
6. Favourable	10
7. Unfavourable	11
8. Mechanisms of ageing	12
9. Practical suggestions	13
10. Limitations	14
11. Methods and sources	15
12. Annex: all variants analysed	16

1. Summary

Favourable. Gastroesophageal reflux has a low genetic predisposition: the scores are in the 6.8 (836 thousand variants) and 10.4 percentiles, and the loci load at percentile 12.5. The hiatus hernia is at percentile 16.1 and Barrett's esophagus or adenocarcinoma of the esophagus in 14.1 (the isolated adenocarcinoma is typical, at 45.3). Gastric cancer has a loci load at percentile 8.4, without the risk allele of MUC1 or PLCE1. In *H. pylori*, FCGR2A is not at risk and TNF -308 is G/G. The CYP2C19 is *1/*1 (normal metabolizer of proton pump inhibitors).

Unfavourable / mixed. The PSCA rs2294008 is T/T: more diffuse gastric cancer (~1.3× per allele), but less duodenal ulcer. **ABO rs505922 T/T** suggests group O, which has more duodenal ulcer but less gastric cancer. In *H. pylori* it has 1 allele of **TLR1** associated with more seropositivity (OR ~1.4) and is heterozygous in **IL1B -511** (more inflammation in infection). It also has 1 risk allele in PRKAA1 and CDH1 -160. Some Barrett loci are at risk homozygosity (BARX1 and FOXP1), but Barrett's genomic score is low. The oesophagous cancer score (14 variants) is at percentile 84.8, and has little zero alcohol. Peptic ulcer (51.2), duodenal ulcer (30.8) and gastritis (67.3) are typical.

What does it mean? The esophagus is of low genetic risk. The stomach has a point to watch: if there is *H. pylori*, PSCA T/T and IL1B add up to the effect of the bacterium on the risk of gastritis and gastric cancer. Therefore, the most impactful step is to know if there is *H. pylori* and eradicate it if there is, with a scheme compatible with rapamycin.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 19 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 17 of the chip, 2 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 2 SNPs needed conversion to the complementary strand and 1 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 10 PGS Catalog scores calculated locally with 2,670,142 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 200 loci, 135 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In all scores, a higher percentile = more risk.

Reflux and esophagus

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Esophagitis, reflux and related diseases	PGS002063 (Privé 2022, LDpred2, 836 thousand variants)	820,027 836,413 (weighted coverage 97%)	6.8	LESS LIKELY	favourable
Esophagitis, reflux and related diseases	PGS001851 (Privé 2022, 13.9 thousand variants)	13,437 13,855 (weighted coverage 96%)	10.4	LESS LIKELY	favourable
Hiatal hernia	PGS000939 (Tanigawa 2022, 2553 variants)	2,217 2,553 (weighted coverage 88%)	16.1	LESS LIKELY	favourable
Barrett's esophagus or esophageal adenocarcinoma	PGS003387 (Namba 2022, 602 thousand variants)	522,372 601,980 (weighted coverage 82%)	14.1	LESS LIKELY	favourable
Esophageal adenocarcinoma	PGS003388 (Namba 2022, 357 thousand variants)	275,813 356,743 (weighted coverage 69%)	45.3	TYPICAL	typical
Oesophageal cancer	PGS002298 (14 variants)	13 14 (weighted coverage 98%)	84.8	MORE LIKELY	unfavourable

Stomach and duodenum

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Peptic ulcer	PGS012562 (Li 2024, 8 variants)	8 8 (weighted coverage 100%)	51.2	TYPICAL	typical
Duodenal ulcer	PGS001390 (Tanigawa 2022, 220 variants)	173 220 (weighted coverage 82%)	30.8	TYPICAL	typical
Gastritis and duodenitis	PGS004470 (Jung 2024, LDpred2, 1.06 million variants)	1,036,079 1,059,939 (weighted coverage 97%)	67.3	TYPICAL	typical
Gastric cancer	PGS002299 (Choi 2020, 3 variants)	3 3 (weighted coverage 100%)	16.4	LESS LIKELY	favourable (low reliability)

- **Gastric cancer:** Only 3 variants: low reliability.

4. Key variants by topic

Variants with described effect. The judgment refers to the esophagus, stomach and duodenum; in CYP2C19 it refers to the response to proton pump inhibitors.

Gastroesophageal reflux, Barrett's esophagus and hiatus hernia

The reflux loci and Barrett's esophagus are mainly transcription factors of the development of the esophagus and gastroesophageal junction (FOXF1, BARX1, FOXP1, GDF7). Barrett is a metaplasia: the epithelium of the esophagus changes due to chronic reflux.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
FOXF1	rs9936833		C/T	chip	C (1 copy/s)	unfavourable (light)	Transcription factor of oesophagus and stomach development (Barrett and adenocarcinoma of the esophagus) [GWAS Catalog: Barrett's esophagus, allele C, OR 1.14, p=3E-10, PMID 22961001]	Moderate
BARX1	rs11789015		A/A	chip	A (2 copy/s)	unfavourable	Development of the esophagus and gastroesophageal junction (Barrett) [GWAS Catalog: Digestive system disease (Barrett's esophagus and esophageal, allele A, OR 1.20, p=1E-9, PMID 24121790]	Moderate
FOXP1	rs2687201		A/A	chip	T (2 copy/s)	unfavourable	Development of the esophagus (Barrett) [GWAS Catalog: Digestive system disease (Barrett's esophagus and esophageal, allele T, OR 1.18, p=5E-9, PMID 24121790]	Moderate
GDF7	rs3072		T/C	imputation DR2=0.98	C (1 copy/s)	unfavourable (light)	BMP signaling in the esophagus epithelium (Barrett) [GWAS Catalog: Barrett's esophagus, T allele, beta 0.113 unit decrease, p=9E-13, PMID 34187846] (the reported T allele reduces the trait; here the effect allele is the opposite, C)	Moderate
LINC00208	rs10419226		G/G	chip	A (0 copy/s)	favourable	Gastroesophageal reflux locus and Barrett (19p13) [GWAS Catalog: Digestive system disease (Barrett's esophagus and esophageal, allele A, OR 1.18, p=4E-10, PMID 24121790]	Weak

Helicobacter pylori: infection and inflammatory response

Helicobacter pylori is the main cause of peptic ulcer and gastric cancer. Host genes affect the likelihood of infection (TLR1, FCGR2A), inflammation strength (IL1B, TNF) and adhesion of bacteria to the mucosa (FUT2).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
TLR1/ TLR6/ TLR10	rs10004195		T/A	chip	A (1 copy/s)	unfavourable (light)	Toll-like receptors that recognize H. pylori: the risk allele is associated with more seropositivity (more likely or more persistent infection) [GWAS Catalog: Helicobacter pylori serologic status, allele A, OR 1.43, p=1E-18, PMID 23652523]	Moderate

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
FCGR2A	rs368433		T/T	imputation DR2=0.90	C (0 copy/s)	favourable	Fc-gamma receptor: associated with seropositivity for H. pylori [GWAS Catalog: Helicobacter pylori serologic status, allele C, OR 1.37, p=2E-8, PMID 23652523]	Weak
IL1B	rs16944	-511	A/G	chip	A (1 copy/s)	unfavourable (light)	-511C>T (gene T allele = A in strand +): plus IL-1 β in the mucosa and more acid suppression in H. pylori infection; associated with atrophic gastritis and gastric cancer (moderate and inconsistent effect)	Weak
TNF	rs1800629	-308	G/G	chip	A (0 copy/s)	favourable	-308G>A: allele A increases TNF- α ; weak association with gastritis and gastric cancer in H. pylori infection	Weak
FUT2	rs601338	W154X	G/A	chip	A (1 copy/s)	neutral	W154X: Allele A (non-secretor in homozygous) removes Lewis b antigens from the mucosa, where H. pylori is fixed; non-secretors may have less adhesion of the bacterium	Weak

Peptic ulcer and gastric mucosa

Mucus barrier (MUC1) and loci shared by peptic ulcer and gastric cancer. In PSCA and ABO the same allele protects from one disease and increases the other, so judgment is mixed.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PSCA	rs2294008		T/T	chip	T (2 copy/s)	mixed	Prostate stem cells antigen expressed in the stomach: T allele increases diffuse gastric cancer, but reduces duodenal ulcer	Moderate
ABO	rs505922		T/T	chip	T (2 copy/s)	mixed	Group O marker (T allele): group O is associated with more duodenal ulcer (~1.3 $\times$), but less gastric cancer; T/T suggests group O	Moderate
MUC1	rs4072037		C/C	chip	T (0 copy/s)	favourable	Mucin 1: splicing variant associated with ulcer and gastric cancer [GWAS Catalog: Non-cardia gastric cancer, allele T, OR 1.35, p=6E-17, PMID 26701879]	Moderate

Gastric cancer

Loci GWAS of gastric cancer and E-cadherin (CDH1). The severe hereditary variants of CDH1 are rare and do not read on the chip.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PRKAA1	rs13361707		C/T	chip	C (1 copy/s)	unfavourable (light)	Gastric cancer locus 5p13 (AMPK α 1) [GWAS Catalog: Gastric cancer, C allele, OR 1.18, p=1E-21, PMID 30281874]	Moderate
PLCE1	rs2274223		A/A	chip	G (0 copy/s)	favourable	Phospholipase C epsilon: gastric cancer of cardia and esophagus [GWAS Catalog: Esophageal cancer, G allele, OR 1.34, p=4E-20, PMID 21642993]	Weak

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CDH1	rs16260		C/A	chip	A (1 copy/s)	unfavourable (light)	-160C>A: allele A reduces the transcription of E-cadherin; small and inconsistent effect on gastric cancer	Weak

Response to proton pump inhibitors (PPI)

CYP2C19 metabolizes omeprazole, esomeprazole, lansoprazole and pantoprazole. The CPIC guideline uses the genotype to adjust the dose.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CYP2C19	rs4244285	*2	G/G	chip	A (0 copy/s)	favourable	*2: loss of function; slow metabolizers have more exposure to PPI (omeprazole, pantoprazole)	Strong
CYP2C19	rs4986893	*3	G/G	chip	A (0 copy/s)	favourable	*3: loss of function	Strong
CYP2C19	rs12248560	*17	C/C	chip	T (0 copy/s)	favourable	*17: more activity; fast metabolizers have less exposure and less efficacy of PPI	Strong

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of each disease, with each allele oriented towards the risk, compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Gastroesophageal reflux	7 (6)	2 of 12	6	12.5	favourable
Barrett's esophagus / adenocarcinoma of the esophagus	51 (41)	31 of 82	41	25.0	typical
Peptic ulcer (gastric and duodenal)	94 (56)	48 of 112	54	33.3	typical
Gastric cancer	48 (32)	25 of 64	32	8.4	favourable

6. Favourable

- **Gastroesophageal reflux:** scores at the 6.8 and 10.4 percentiles; loci load at percentile 12.5. **Moderate.**
- **Hiatal hernia:** percentile 16.1. **Moderate.**
- **Barrett's esophagus or esophageal adenocarcinoma:** percentile 14.1 (602 thousand variants); loci load at 25.0. LINC00208 without risk allele. **Moderate.**
- **Gastric cancer:** loci load at percentile 8.4; MUC1 C/C and PLCE1 A/A without risk alleles. The 3 variant score (percentile 16.4) has low reliability. **Moderate.**
- **H. pylori:** FCGR2A without seropositivity allele; TNF -308 G/G (without the allele plus TNF). **Weak.**
- **CYP2C19 *1/*1** (without *2, *3 or *17): normal metabolizer of PPI. **Strong.**

7. Unfavourable

- **PSCA rs2294008 T/T**: more diffuse gastric cancer ($\sim 1.3\times$ per allele), but less duodenal ulcer. Moderate (mixed).
- **ABO rs505922 T/T** (probable group O): more duodenal ulcer ($\sim 1.3\times$), but less gastric cancer. Moderate (mixed).
- **TLR1/TLR6/TLR10 rs10004195 T/A**: 1 allele of more seropositivity for *H. pylori* (OR ~ 1.4). **Moderate.**
- **IL1B -511 heterozygote** (A/G): plus IL-1 β in the infected mucosa, atrophic gastritis and hypochloridry. Weakness (inconsistent effect).
- **Loci do Barrett**: BARX1 A/A and FOXP1 A/A in risk homozygosity FOXP1 and GDF7 with 1 allele. Barrett's genomic score is low, so the set is attenuated. **Weak.**
- **Oesophageal cancer** (14 variant score): percentile 84.8. Many of these loci act through alcohol and tobacco, so they have little zero alcohol. **Weak.**
- **PRKAA1 C/T and CDH1 -160 C/A**: 1 risk allele in gastric cancer. **Weak.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Dysbiosis	H. pylori: TLR1 with 1 allele of more seropositivity; FCGR2A without risk; FUT2 secretor (typical bacterial adhesion)	Unfavourable (mild)
Chronic Inflammation	IL1B -511 heterozygous (plus IL-1β in gastritis by H. pylori); TNF -308 G/G	Neutral to unfavourable (mild)
Intestinal Barrier dysfunction	Gastric mucus barrier: MUC1 without the risk allele; PSCA T/T	Mixed
Aberrant Transcriptional Reprogramming	Barrett metaplasia: BARX1 and FOXP1 in homozygosity, but Barrett's genomic score at percentile 14.1	Neutral
Somatic Mutation Accumulation	Gastric cancer: loci load at percentile 8.4, but PSCA T/T (diffuse type)	Mixed
Inflammaging	Gastritis and duodenitis at percentile 67.3 (typical)	Neutral

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **H. pylori test** (breath test with urea or antigen in the feces), if it was never done. Do the test without PPI in the previous 2 weeks. PSCA T/T and IL1B add to the bacteria in the risk of gastritis and gastric cancer, therefore eradicating an infection is the step with more impact. *Suggestion (not applied)*.
2. **If necessary:** CYP2C19 *1/*1 eliminates PPI faster than a slow metabolizer. The CPIC guideline suggests increasing the dose of PPI in 50–100% in the eradication of H. pylori and erosive esophagitis. The critical point is clarithromycin, a strong inhibitor of CYP3A4 that greatly increases the level of sirolimus. Therefore, the best is a scheme without clarithromycin (quadruple with bismuth), outside the rapamycin window. Confirm 4 eradication weeks later. *Suggestion (not applied)*.
3. **NSAIDs:** with group O and PSCA, paracetamol remains the 1.a line, as in the plan. If a NSAID is necessary, use the minimum dose and for a short time, with gastric protection if it is more than a few days. *Suggestion (not applied)*.
4. **Reflux:** The genetic predisposition is low. If heartburn or nausea appear, the most likely is retatrutide, which delays gastric emptying. Keep the last meal ≥ 3 h before bedtime (already in the plan). *Suggestion (not applied)*.
5. **Endoscopy:** genes do not justify it by itself. It only makes sense if there are symptoms of alarm or chronic reflux. *Suggestion (not applied)*.
6. **Protective diet of the stomach (already in the plan):** salt < 5 g/day (salt is a risk factor for gastric cancer), avoiding processed, smoked and carbonized meat, and daily crucifers. Broccoli shoots (sulforaphane) have evidence to reduce colonization by H. pylori. High fiber: the more, the better. *Suggestion (not applied)*.
7. **PPI only with indication and for the minimum time:** The chronic use reduces the absorption of B12 and magnesium and stirs in the microbiome. The B12 of 1000 µg of the plan covers the B12. *Suggestion (not applied)*.
8. **Zero alcohol (already in progress):** maintains low risk of esophageal cancer, which is the only high score in this report. *Suggestion (not applied)*.

Not recommended: eradication regimens with clarithromycin while taking rapamycin PPI in continuous without indication.

10. Limitations

- A genotyping chip with imputation does not read the severe hereditary variants of CDH1 (diffuse hereditary gastric cancer), the rare variants of other gastric cancer syndromes, or the infection itself. *H. pylori* is only known with one test.
- The reflux and ulcer scores are based on diagnoses of health records and self-declaration, so they have noise.
- The gastric cancer score has only 3 variants and the esophageal cancer score mixes both histological types. Both have low reliability.
- The blood group is deduced from a marker (rs505922) and should be confirmed by typing.
- The load of loci GWAS sums SNPs in linkage imbalance. It is a descriptive complement and does not replace the polygenic scores.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The alleles of each location are read from the Imputed VCF. The published effect allele is compared with those alleles and, if necessary, converted to the complementary strand (only in non-palindromic SNPs). In palindromic SNPs, strand + (GWAS Catalog standard and the harmonized files of PGS Catalog) is assumed. In the loci of GWAS Catalog with protective effect, the allele presented is that of risk.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog: PGS002063 and PGS001851 (Privé 2022), PGS000939 and PGS001390 (Tanigawa 2022), PGS003387 and PGS003388 (Namba 2022), PGS002298 and PGS002299 (Choi 2020), PGS012562 (Li 2024) and PGS004470 (Jung 2024). Score = Δ coefficient $\times$ dose of effect allele. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog (associations with $p \leq 5 \times 10^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** GWAS Catalog, ClinVar, dbSNP/Ensembl and reference literature: Ong 2022 (reflux), Levine 2013 and Gharahkhani 2016 (Barrett), Mayerle 2013 (*H. pylori*), El-Omar 2000 (IL1B), Wu 2021 (peptic ulcer), Sakamoto 2008 (PSCA) and the guideline CPIC of 2021 for CYP2C19 and PPI. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Gastroesophageal reflux, Barrett's esophagus and hiatus hernia	FOXF1	rs9936833	C/T	chip	C	1	strand +	1 copy of risk allele
Gastroesophageal reflux, Barrett's esophagus and hiatus hernia	BARX1	rs11789015	A/A	chip	A	2	strand +	2 risk allele copies
Gastroesophageal reflux, Barrett's esophagus and hiatus hernia	FOXP1	rs2687201	A/A	chip	T	2	complement (strand)	2 risk allele copies
Gastroesophageal reflux, Barrett's esophagus and hiatus hernia	GDF7	rs3072	T/C	imputation DR2=0.98	C	1	strand +	1 copy of risk allele
Gastroesophageal reflux, Barrett's esophagus and hiatus hernia	LINC00208	rs10419226	G/G	chip	A	0	complement (strand)	without the risk allele
Helicobacter pylori: infection and inflammatory response	TLR1/TLR6/TLR10	rs10004195	T/A	chip	A	1	strand + (palindromic; assumed +)	1 copy of risk allele
Helicobacter pylori: infection and inflammatory response	FCGR2A	rs368433	T/T	imputation DR2=0.90	C	0	strand +	without the risk allele
Helicobacter pylori: infection and inflammatory response	IL1B	rs16944	A/G	chip	A	1	strand +	1 copy of risk allele
Helicobacter pylori: infection and inflammatory response	TNF	rs1800629	G/G	chip	A	0	strand +	without the risk allele
Helicobacter pylori: infection and inflammatory response	FUT2	rs601338	G/A	chip	A	1	strand +	G/A: secretor
Peptic ulcer and gastric mucosa	PSCA	rs2294008	T/T	chip	T	2	strand +	T/T: more diffuse gastric cancer (~1.3x per allele) and less duodenal ulcer
Peptic ulcer and gastric mucosa	ABO	rs505922	T/T	chip	T	2	strand +	T/T: probable group O (more duodenal ulcer, less gastric cancer)
Peptic ulcer and gastric mucosa	MUC1	rs4072037	C/C	chip	T	0	strand +	without the risk allele
Gastric cancer	PRKAA1	rs13361707	C/T	chip	C	1	strand +	1 copy of risk allele
Gastric cancer	PLCE1	rs2274223	A/A	chip	G	0	strand +	without the risk allele
Gastric cancer	CDH1	rs16260	C/A	chip	A	1	strand +	1 copy of risk allele
Response to proton pump inhibitors (PPI)	CYP2C19	rs4244285	G/G	chip	A	0	strand +	without the risk allele

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Response to proton pump inhibitors (PPI)	CYP2C19	rs4986893	G/G	chip	A	0	strand +	without the risk allele
Response to proton pump inhibitors (PPI)	CYP2C19	rs12248560	C/C	chip	T	0	strand +	without the risk allele